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The Rural Postman Problem on mixed graphs with turn penalties

Angel Corberán[a,*], Rafael Martí[a], Eulalia Martínez[b], David Soler[b]

[a]*Facultad de Matematicas, Departamento de Estadística e Investigación Operativa, Universitat de València, Dr. Moliner 50, 46100 Burjassot, València, Spain*

[b]*Departamento de Matemática Aplicada, Universidad Politécnica de Valencia, Camino de Vera s/n, 46022, Valencia, Spain*

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Abstract

In this paper we deal with a problem which generalizes the Rural Postman Problem defined on a mixed graph (MRPP). The generalization consists of associating a non-negative penalty to every turn as well as considering the existence of forbidden turns. This new problem fits real-world situations more closely than other simpler problems. A solution tour must traverse all the requiring service arcs and edges of the graph while not making forbidden turns. Its total cost will be the sum of the costs of the traversed arcs and edges together with the penalties associated with the turns done. The Mixed Rural Postman Problem with Turn Penalties (MRPPTP) consists of finding such a tour with a total minimum cost. We show that the new problem is NP-hard, even in some particular cases. In order to solve the MRPPTP, a polynomial transformation of the problem into the Asymmetric Traveling Salesman Problem (ATSP) can be done and then apply heuristics and exact methods for the ATSP. This transformation is summarized here and a specific heuristic algorithm, based on two recent heuristics for the MRPP, is also presented. Extensive computational experiments with more than 250 instances establish the effectiveness of our procedures.

Scope and purpose

Arc Routing Problems consist, basically, of finding a set of routes over the arcs (and/or edges) of a graph while satisfying certain conditions. Such problems arise in a variety of practical contexts, such as garbage collection, snow removal, electrical or gas mains inspection, mail delivery, etc. In most applications, there are some restrictions on the vehicle to traverse some streets on a specified way and the problem needs to be modeled with a mixed graph. Moreover, restrictions on the turns can be also considered. In this paper, we focus with such a problem: the Mixed Rural Postman Problem with Turn Penalties. This is an NP-hard problem for which two different solution procedures are presented. © 2002 Elsevier Science Ltd. All rights reserved.

Keywords: Arc routing; Rural postman problem; Turn penalties; Metaheuristics

* Corresponding author. Tel.: + 34-6386-4306; fax: + 34-6386-4735.
E-mail address: angel.corberan@uv.es (A. Corberán).

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1. Introduction

Arc Routing Problems consist, basically, of finding a set of routes over the arcs (and/or edges) of a graph while satisfying certain restrictions. There are many real situations in which such a set of routes is required: collection or delivery of goods, garbage collection, snow removal, electrical or gas mains inspection, mail delivery, etc. For this reason, Arc Routing Problems have been extensively studied in recent years. The excellent papers by Assad and Golden [1], Eiselt et al. [2,3] and the very recent book edited by Dror [4] summarize the state of the art and real-life applications of these problems. In most applications, there are some restrictions on the vehicle to traverse some streets on a specified way and the problem needs to be modeled with a mixed graph, in which one-way streets are represented by arcs and two-way streets by edges.

In this paper, we focus on a generalization of the Rural Postman Problem defined on a mixed graph (MRPP). The MRPP may be defined as follows. Given a strongly connected mixed graph $G = (V, E, A)$ with vertex set V, edge set E, arc set A, a cost c_e for each $e \in E \cup A$, a set $E_R \subseteq E$ of *required edges* and a set $A_R \subseteq A$ of *required arcs*, the MRPP is the problem of finding a minimum cost closed walk which passes through each $e \in R = E_R \cup A_R$ at least once.

Note that the MRPP reduces to the *Chinese Postman Problem* (CPP) when $A = \emptyset$ and $R = E$, to the *Directed Chinese Postman Problem* (DCPP) when $E = \emptyset$ and $R = A$ and to the *Mixed Chinese Postman Problem* (MCPP) when $R = E \cup A$ and $A \neq \emptyset \neq E$. Furthermore, the *Rural Postman Problem* (RPP), the *Directed Rural Postman Problem* (DRPP) and the *Stacker Crane Problem* (SCP) are also particular cases of the MRPP when $A = \emptyset$ and $R \subset E$, $E = \emptyset$ and $R \subset A$ and when $A \neq \emptyset \neq E$ and $R = A$, respectively.

Different approaches to the MRPP can be found in the recent papers by Laporte [5], Corberán et al. [6] and Corberán et al. [7]. The first paper describes a tranformation of the MRPP into an Asymmetric Traveling Salesman Problem (ATSP). Then, exact and heuristic algorithms for the ATSP (see, for example, Laporte [8]) can be applied to the resolution of the MRPP. The second paper presents both a constructive and a tabu search algorithm that produce, in general, very good solutions. Finally, the study of the polyhedron associated with the MRPP is one of the subjects of the third paper.

However, when a real-life problem is modelled and solved as a MRPP, or as a particular case of it, it should be noted that we are assuming that all turns are allowed and that they are not time-consuming. While some turns can be considered irrelevant, others are more time-consuming and/or dangerous in their execution, specially for trucks. Furthermore, most U-turns may even be forbidden. Therefore, the MRPP solution could be infeasible if the traffic signals are respected, specially inside a city. Even when considering a tour on foot (a postman's tour, for example), on reaching crossing v from sidewalk a (see Fig. 1), it is evident that it does not take the same time to go on along sidewalk b (left turn) than to go on along sidewalk e, for which it might be necessary to cross up to two traffic lights.

Several authors have taken turn penalties into account in order to solve real problems at a heuristic level: Roy and Rousseau [9] in the design of postman routes, Bodin and Kursh [10,11] for routing street sweepers and Bodin et al. [12] for vehicle routing in garbage collection. See also McBride [13], where several ideas are presented in order to avoid left and U-turns on a given route. Benavent and Soler [14] also consider the existence of forbidden turns when searching for an optimal solution of the Directed Rural Postman Problem with Turn Penalties (DRPPTP). In this

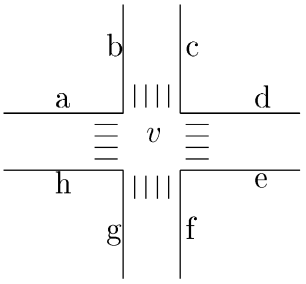


Fig. 1. Street crossing v: not all the alternatives to continue when a postman arrives at v from sidewalk a are equally time-consuming.

problem, a non-negative penalty is associated with every turn. Each time the vehicle goes from one arc to another, the solution cost is increased by the corresponding penalty. Forbidden turns have an infinite penalty. These authors propose solving this problem optimally by transforming it into an ATSP. Several heuristics to solve the DRPPTP are also presented and tested in 30 instances, six of them corresponding to real city areas. Particular cases of the same problem in which all U-turns are forbidden and where there is no penalty for the remaining tours can be found, among others, in Benavent and Soler [15] and Soler [16].

Following the results in Laporte [5] and in Benavent and Soler [14], we consider here the Mixed Rural Postman Problem with Turn Penalties (MRPPTP). This problem generalizes all the previously presented Arc Routing Problems but also takes into account turn penalties and forbidden turns. Basically, in this paper we discuss first its transformation into an ATSP and we report some computational experience on its resolution by using both an exact and a heuristic method for the ATSP. Second, we present a heuristic algorithm that applies directly to the MRPPTP and that, as the computational results show, produces very good solutions. More specifically: after introducing notation and some definitions in Section 2, we present in Section 3 a result regarding the complexity of the MRPPTP and its transformation into an ATSP. A heuristic algorithm for the MRPPTP is described in Section 4 and finally, in Section 5, we present the computational results obtained with our heuristic as well as with an exact and with a heuristic procedure for the ATSP on a set of 279 randomly generated instances. In all of them, U-turns are forbidden and the penalties assigned to the remaining turns depend on their type (straight ahead, to the right or to the left). Instances tested have up to 200 vertices, 440 required arcs and 100 required edges. As in the computational experience carried out by Laporte [5], our results show that the transformation of the MRPPTP into an ATSP works well in instances with few required edges, while the behavior of our heuristic does not depend on this and produces solutions with an average deviation from the optimal solution about 1%.

2. Some definitions and notation

Given a mixed graph $G = (V, E, A)$, in order to simplify the notation, an element of $E \cup A$ will be called a *link*. Then, each pair of links $a = (u, v), b = (v, w) \in E \cup A$ has a turn associated in v: the one

made going from a to b, which will be denoted $[ab]$. Moreover, if $a, b \in E$, this pair has another turn associated in v: the one made going from b to a, denoted $[ba]$. Each edge e incident to v has a U-turn associated in v that, if necessary, will be denoted by $[eve]$. In the same way, when $e, e' \in E$ represent two edges in parallel between the same vertices u and v, in order to distinguish the turns at u and v, we will denote them by $[eue']$ and $[eve']$, respectively. Each link $a \in E \cup A$ has an associated cost $c_a \geqslant 0$ and each turn $[ab]$ in G has an associated penalty $p_{[ab]} \geqslant 0$ ($p_{[ab]} = +\infty$ iff $[ab]$ is a forbidden turn).

Given $a = (u, v)$, $b = (s, t) \in E \cup A$, a *$v - s$ feasible chain from a to b* is an alternate sequence of links and turns $C = \{a_1, [a_1 a_2], a_2, \ldots, [a_{r-1} a_r], a_r, [a_r b]\}$, where $a_1 = a$, satisfying that $\{a_1, \ldots, a_r, b\}$ is a chain in G, turn $[a_r b]$ is allowed in G and, if $r > 1$, $[a_i a_{i+1}]$, $i = 1, \ldots, r-1$, are allowed turns in G. Note that, for convenience, a feasible chain begins at a link and ends at a turn and that, if all links involving C are edges, this feasible chain must be traversed only in the specified direction. The cost of a feasible chain C is defined as $c(C) = \sum_{i=1}^{r} (c_{a_i} + p_{[a_i a_{i+1}]})$, where $a_{r+1} = b$.

Given $a = (u, v), b = (s, t) \in E \cup A$, a *shortest $v - s$ feasible chain from a to b* is a $v - s$ feasible chain C from a to b of minimum cost $c(C)$. This feasible chain will be denoted by *s.f.c.*(v^a, s^b). We use shortest feasible chains (s.f.c.) instead of shortest feasible paths, because a *s.f.c.* can pass through the same vertex more than once in G. A $v - s$ feasible chain from a to b is *closed* if $a = b$ and $s \neq v$ ($b = (s, v)$).

Now, we can formally define the MRPPTP as follows:

Let $G = (V, E, A)$ be a mixed graph with associated link costs and turn penalties as defined above. Given a non-empty subset R of $E \cup A$, find a minimum cost feasible closed chain in G, containing each link of R at least once.

Links in R will be called *required links* and a feasible closed chain containing each required link at least once will be called a *MRPPTP tour*.

As was pointed out in Section 1, this problem generalizes the MRPP and thus all the Arc Routing Problems presented there. To see this, it is sufficient to consider all turns as allowed turns with zero penalties. The particular case of the MRPPTP in which $E = \emptyset$ has been recently studied by Benavent and Soler [14]. As will be shown in the computational results, this particular problem seems to be the easiest case. The problem presented in Roy and Rousseau [9] can also be considered as a particular case of the MRPPTP in which $A = \emptyset$, G is Eulerian, $R = E$, $p_{[ea]} = p_{[ae]} < +\infty$ for every $[ea]$ turn in G and all the turn penalties are small enough in comparison with the edge costs, so that the optimal solution passes through each edge exactly once. Furthermore, the problems with turn penalties introduced in Bodin and Kursh [10,11] and Bodin et al. [12] are particular cases of the MRPPTP in which all turns are allowed and $A = \emptyset$ or $E = \emptyset$. Basically, their algorithms first look for a heuristic solution to the Rural Postman Problem on an undirected, or directed, graph. Note that with respect to this solution each vertex has even degree. Then, in the case of a directed graph (for example), an Assignment Problem (AP) in which an arc entering at any vertex v is assigned to an arc leaving it (with cost that of the associated turn penalty) is solved. From the AP solution, several circuits are obtained and will be merged to obtain a final traversal of the RPP tour. The above heuristics cannot be directly applied when we consider the existence of forbidden turns, because although very high penalties were associated to the forbidden turns, they cannot be guaranteed to produce a MRPPTP tour. Consider, for example, the undirected graph of Fig. 2(a), in which all edges are required and have cost 1, all U-turns and turns

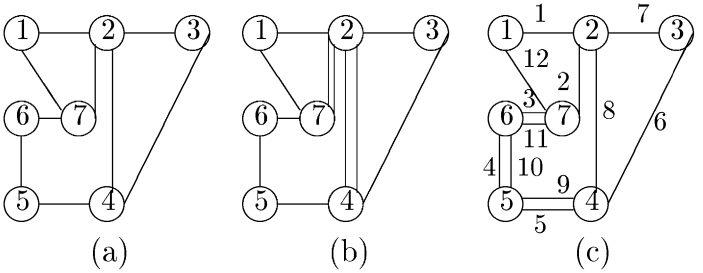


Fig. 2.

[(7, 2)(2, 4)] and [(4, 2)(2, 7)] are forbidden, and the remaining turns are allowed with penalty zero. In this case, the MRPP reduces to the well-known CPP. Fig. 2(b) shows the CPP solution, which does not admit an Eulerian tour without violating at least one forbidden turn at vertex 2. Fig. 2(c) represents an optimal MRPPTP tour, where edges have been numbered according to their traversing order in the tour.

3. Complexity results and MRPPTP transformation

In this section, we will discuss how to transform the MRPPTP into an ATSP. This transformation makes it possible to solve the MRPPTP by using the existing exact and heuristic algorithms for the ATSP. The reason for this transformation is that the MRPPTP is an NP-hard problem, even in the particular cases where the corresponding Arc Routing Problems without taking into account turn penalties and forbidden turns are polynomially solvable (the Undirected and DCPP). In fact, the MRPPTP is NP-hard even in the following cases:

- $E = \emptyset, R = A$ and G is Eulerian.
- $A = \emptyset, R = E$, G is Eulerian and $p_{[ab]} = p_{[ba]}$ for every $[ab]$ turn in G.

The first result has been discussed in Benavent and Soler [14]. We will show the second one in the following theorem which transforms the Symmetric TSP into this case.

Theorem 1. *The TSP defined in a simple undirected graph $G = (V, E)$ can be transformed polynomially into a MRPPTP defined in an Eulerian graph $H = (V', E')$ with $R = E'$ and $p_{[ea]} = p_{[ae]}$ for every $[ea]$ turn in H.*

Proof. Given the simple graph $G = (V, E)$ in which a TSP is defined, we construct a graph $H = (V', E')$ as follows (see Fig. 3):

- If $V = \{i\}_{i=1}^{n}$, then $V' = \{i\}_{i=1}^{2n} \cup \{u\}$,
- $E' = \{(i, n+i)\}_{i=1}^{n} \cup \{(i, u)\}_{i=1}^{n} \cup \{(u, n+i)\}_{i=1}^{n}$.

(Vertex $n + i$ can be considered a duplication of vertex i, $i = 1, \ldots, n$ and vertex u is a new vertex adjacent to the remaining ones.)

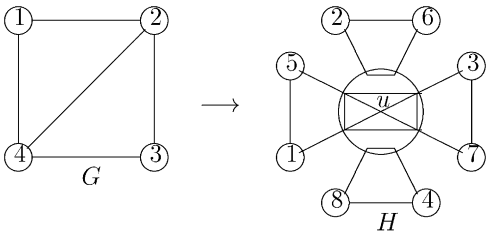


Fig. 3. Transforming a TSP in G into a MRPPTP in H.

We associate a very high-cost M to all edges in E' and the following penalties to the turns in H:

- $p_{[(u,i)(i,n+i)]} = p_{[(u,n+i)(n+i,i)]} = 0 \ \forall i = 1, \ldots, n.$
- $p_{[(i,u)(u,j)]} = p_{[(i,u)(u,n+j)]} = p_{[(n+i,u)(u,n+j)]} = c_{ij}$ if $(i, j) \in E$, where c_{ij} is the cost in graph G of edge (i, j).
- $p_{[(i,u)(u,j)]} = p_{[(i,u)(u,n+j)]} = p_{[(n+i,u)(u,n+j)]} = +\infty$ (forbidden turn) if $(i, j) \notin E$ or $i = j$.
- $p_{[eve]} = +\infty$ for every $[eve]$ U-turn in H.
- $p_{[ea]} = p_{[ae]}$ for every $[ea]$ turn in H.

The TSP has a solution in G if and only if the MRPPTP has a solution in H traversing each edge of E' exactly once. It is easy to transform such a solution of the MRPPTP in H into a solution of the TSP in G and vice versa. To do this, it is sufficient to associate the edge (i, j) in G traversed from i to j with the following feasible chain in H: $\{(i, n+i)[(i, n+i)(n+i, u)](n+i, u)[(n+i, u)(u, j)](u, j)[(u, j)(j, n+j)]\}$. Moreover, when a solution of this type exists for the MRPPTP in H, its optimal solution is also of this type due to the high cost associated to edges in E'. □

Fig. 3 shows an example of the above transformation. Assume that all edges in G have cost 1. Then, allowed turns have penalty 1 in vertex u and penalty 0 in the remaining vertices. Turns drawn with lines in vertex u are forbidden in both directions as well as all U-turns. The optimal MRPPTP tour in H is: $\{(1, 5)(5, u)(u, 2)(2, 6)(6, u)(u, 3)(3, 7)(7, u)(u, 4)(4, 8)(8, u)(u, 1)\}$, in which turns have been omitted in order to simplify the expression. From this tour we get the optimal TSP solution in G: $\{(1, 2),(2, 3),(3, 4),(4, 1)\}$.

In order to transform the MRPPTP into an ATSP we need to compute shortest feasible chains in a graph with turn penalties and forbidden turns. This is not an easy task if we try to work directly on the original graph. The method we use here follows the ideas proposed in Kirby and Potts [17] (see also Roy and Rousseau [9] and Eiselt et al. [3]): Let $G = (V, E, A)$ be a mixed graph in which turn penalties and forbidden turns have been defined. An associated directed graph, D_G, is defined as follows:

- For each edge $e = (u, v) \in E$, D_G has four vertices, denoted by u_i^e, u_f^e, v_i^e and v_f^e, and two non-adjacent arcs with opposite directions: (u_i^e, v_f^e) and (v_i^e, u_f^e), both with the same cost c_e. Note that i and f distinguish if a vertex is the initial or the final vertex, respectively, of a given arc.
- For each arc $a = (u, v) \in A$, D_G has two vertices, denoted by u_i^a and v_f^a, and an arc (u_i^a, v_f^a) with cost c_a.
- For each allowed turn $[ab]$ in G, D_G has an arc (v_f^a, v_i^b) ((v_f^a, v_i^a) for a U-turn) with cost $p_{[ab]}$.

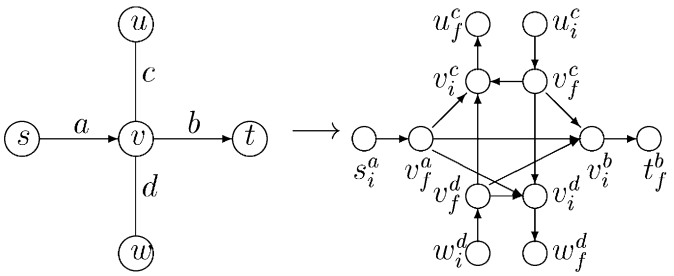


Fig. 4.

Fig. 4 shows an example of the above transformation. Link costs and turn penalties have not been included and all turns in vertex v are allowed.

Given $a = (u, v), b = (s, t) \in E \cup A$, it is easy to see that a $v - s$ feasible chain from a to b in G is in a 1 to 1 correspondence with a chain in D_G from u_i^a to s_i^b, and the transformation of one into another is evident. Hence, the *s.f.c.*(v^a, s^b) in G corresponds to the shortest path from u_i^a to s_i^b in D_G and then, for example, Dijkstra's algorithm to find the shortest paths in D_G can be applied. Since $|V(D_G)| = 4|E| + 2|A|$, this search has polynomial complexity $O((4|E| + 2|A|)^2)$.

Consider now a MRPPTP defined on a mixed graph G, where again R denotes its set of required arcs and edges. A directed graph G', associated with G, will be constructed as follows:

- Associated with each arc $a \in R$, G' has a vertex x_a.
- Associated with each edge $e = (u, v) \in R$, G' have two vertices, u_e and v_e, and two arcs, (u_e, v_e), (v_e, u_e), both with cost $-M$, where M represents a very large positive number.
- For each pair of different edges $a = (u, v), b = (s, t) \in R$, add the following arcs to G' (if the corresponding s.f.c. exists in G):
  - (u_a, s_b) and (s_b, u_a) with costs those of *s.f.c.*(u^a, s^b) and *s.f.c.*(s^b, u^a) in G, respectively.
  - (u_a, t_b) and (t_b, u_a) with costs those of *s.f.c.*(u^a, t^b) and *s.f.c.*(t^b, u^a) in G, respectively.
  - (v_a, s_b) and (s_b, v_a) with costs those of *s.f.c.*(v^a, s^b) and *s.f.c.*(s^b, v^a) in G, respectively.
  - (v_a, t_b) and (t_b, v_a) with costs those of *s.f.c.*(v^a, t^b) and *s.f.c.*(t^b, v^a) in G, respectively.
- For each pair of different arcs $a = (u, v), b = (s, t) \in R$, add arcs (x_a, x_b) and (x_b, x_a) to G', with costs those of *s.f.c.*(v^a, s^b) and *s.f.c.*(t^b, u^a) in G (if they exist), respectively.
- For each pair consisting of an arc $a = (u, v)$ and an edge $b = (s, t)$, both in R, add the following arcs to G' (if the corresponding *s.f.c.* exists in G):
  - (x_a, s_b) and (s_b, x_a) with costs those of *s.f.c.*(v^a, s^b) and *s.f.c.*(s^b, u^a) in G, respectively.
  - (x_a, t_b) and (t_b, x_a) with costs those of *s.f.c.*(v^a, t^b) and *s.f.c.*(t^b, u^a) in G, respectively.

Fig. 5 shows an example of a directed graph G' associated with a mixed graph G. Graph G has one required arc $a = (3, 2)$ and two required edges $b = (1, 4)$ and $c = (3, 5)$. Link costs are shown in the figure and all U-turns are forbidden, going straight ahead (according to the graph drawn) has penalty 0, any right turn has penalty 1 and any left turn has penalty 3. Graph G' has only 5 vertices, x_a, 1_b, 4_b, 3_c, 5_c, and the costs of its arcs are shown in the figure. These costs have been computed in G taking into account the above turn penalties.

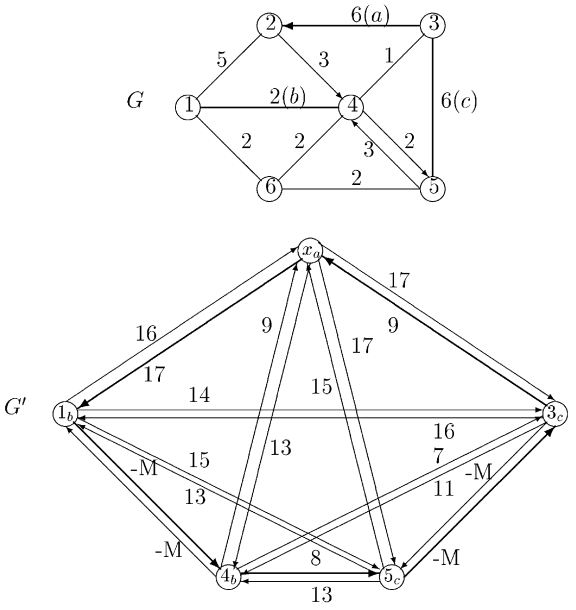


Fig. 5. Graph G with $R = \{a, b, c\}$ and its associated directed graph G'.

Theorem 2. *A MRPPTP defined in* $G = (V, E, A)$, *being* R *the set of required links* ($|R| > 1$), *can be solved by the resolution of an ATSP defined in* G'.

Arguments given in Laporte [5] to justify a similar transformation of the MRPP into an ATSP can also be used here to prove the above result. Note that, although the way to construct graph G' is slightly different to the one in Laporte [5], the resulting graph G' is the same in both constructions except for arc costs. Costs in the tranformation of MRPPTP into ATSP are obtained by computing the shortest feasible chains, which take into account turn penalties and the forbidden turns.

Once the ATSP defined in G' has been solved, its solution H (if it exists) is transformed into a MRPPTP tour in G, T_H, in the following way: each time an arc which is different from (u_e, v_e), with e being any required edge, is traversed in H, T_H traverses the associated *s.f.c.* in G. Note that the cost of T_H is just the cost of H plus kM, whereas k is the number of required edges. Following on with the example in Fig. 5, arcs with a thicker line in G' represent the optimal solution circuit to the ATSP in it: $(x_a, 1_b)$, $(1_b, 4_b)$, $(4_b, 5_c)$, $(5_c, 3_c)$, $(3_c, x_a)$. To obtain the optimal MRPPTP tour in G, note that $(x_a,1_b)$ corresponds in G with the *s.f.c.*$(2^a, 1^b)$: (3, 2)[(3, 2)(2, 1)](2, 1)[(2, 1)(1, 4)], that $(1_b, 4_b)$, $(4_b, 5_c)$ corresponds with the *s.f.c.*$(4^b, 5^c)$: (1, 4)[(1, 4)(4, 5)](4, 5)[(4, 5)(5, 3)], and that $(5_c, 3_c)$, $(3_c, x_a)$ corresponds with the *s.f.c.*$(3^c, 3^a)$: (5, 3)[(5, 3)(3, 2)]. Therefore, the optimal solution to the MRPPTP in G is (removing the interior turns in order to simplify the expression) (3, 2), (2, 1), (1, 4), (4, 5), (5, 3)[(5, 3)(3, 2)] with cost 34 (see Fig. 6).

The optimal MRPPTP tour in the above example is also an optimal tour for the MRPP in G (without taking into account turn penalties) with cost 21. This is a coincidence. In fact, another

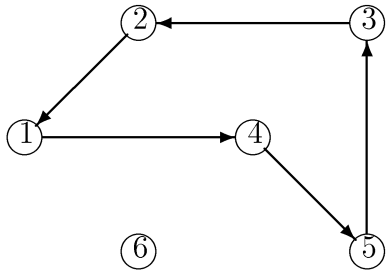


Fig. 6. Optimal solution to the MRPPTP defined in graph G.

optimal MRPP tour in G is $\{(1, 6), (6, 5), (5, 3), (3, 2), (2, 4), (4, 1)\}$, which is a non-optimal MRPPTP tour with cost 37.

In the particular case of the MRPPTP in which $A = \emptyset$ and $p_{[ea]} = p_{[ae]}$ for each $[ea]$ turn in G (as occurs in the problem studied by Roy and Rousseau [9]), we can solve the MRPPTP by transforming it into a Symmetric TSP: Given $e = (u, v)$ and $a = (s, t)$, required edges, and a $v - s$ feasible chain C from e to a, if we delete the cost of edge e in the definition of $c(C)$, the cost of the *s.f.c.*(v^e, s^a) equals that of the *s.f.c.*(s^a, v^e). Therefore, each one of the arcs (i, j) and (j, i) of the directed graph G' constructed above has the same cost. Then, by substituting each pair of arcs (i, j), (j, i) of G' by an edge (i, j) with the same cost, an undirected graph G'' with $2|R|$ vertices is obtained. From the optimal solution to the TSP in G'', an optimal MRPPTP tour in G can be easily obtained.

4. A heuristic for the MRPPTP

As will be seen in the next section, our computational experience shows that the effectiveness of transforming the MRPPTP into an ATSP decreases as the number of required edges increases. In fact, the results obtained on instances with more than 30 required edges are very poor. We present here a heuristic procedure to solve the MRPPTP that consists of three phases. First, several feasible solutions for the MRPP (i.e., without taking into account turn penalties and forbidden turns) are obtained by using the algorithms described in Corberán et al. [6]. Then, these solutions are modified in order to obtain MRPPTP tours and, finally, a third phase tries to improve them. Procedures for last two phases are, basically, extensions to the mixed case of those proposed by Benavent and Soler [14] for the directed case.

4.1. Solving the MRPP

Two heuristics for the MRPP are presented in Corberán et al. [6]: a constructive algorithm partially based on Christofides et al. [18] heuristic for the MCPP, and a tabu search procedure that seeks high-quality solutions in a longer computational time. Computational experiments on graphs with up to 100 vertices and 570 links showed average percentage deviations from a lower bound of 1.87 and 0.47% with the constructive and the tabu search heuristics, respectively. Both

algorithms can be summarized as follows:

Given a strongly connected mixed graph $G = (V, E, A)$ and a subset of required links $R = E_R \cup A_R$, the MRPP consists of finding a minimum cost closed walk traversing each required link at least once (MRPP tour). From G, a simple preprocessing leads to a new graph in which all edges are required and all vertices are incident with required links. Solving the MRPP over the original graph is equivalent to solving it on the resulting graph, that, for notational convenience, will still be denoted by $G = (V, E, A)$, $E = E_R$ and $A = A_R \cup A_{NR}$. The set of required links R induces p connected components in G, which will be called R-connected components, $G_1, G_2, \ldots, G_p$, with vertex sets $V_1, V_2, \ldots, V_p$, respectively. Arcs $(i, j) \in A_{NR}$ such that $i \in V_k$, $j \in V_l$, $k \neq l$ will be called connecting arcs and A_C will denote this arc subset.

A short description of the constructive algorithm follows: Since the tour must connect $G_1, G_2, \ldots, G_p$, the algorithm starts looking for a subset $A' \subseteq A_C$ of connecting arcs such that the graph induced in G by $E \cup A_R \cup A'$ will be connected. This is accomplished by modifying the costs associated to arcs in A_C according to the degree of their incident vertices and computing a shortest spanning tree connecting $G_1, \ldots, G_p$. Then, a minimum cost flow in an auxiliary graph is computed in order to improve the way in which the R-connected components have been connected.

In a second step, a balanced mixed graph is constructed by solving a new minimum cost flow problem with different supplies and demands. The graph is then converted into an even graph by solving a minimum cost matching problem. With this process we obtain a MRPP tour. The algorithm finishes by applying a simple improving procedure.

The tabu search algorithm starts from the MRPP tour obtained with the constructive method. Switches are used as the primary mechanism to move from one solution to another. Specifically, a path containing a connecting arc is replaced with a shortest path. This operation preserves the conditions for symmetry and evenness but may result in disconnected, and therefore infeasible, solutions. Details are given in Corberán et al. [6].

The tabu search basic procedure alternates between an intensification and a diversification phase. In the intensification only improving moves may be performed. In the diversification both non-improving and improving moves may be performed with the aim of escaping from local optima and directing the search to regions in the solution space that have not been explored. The interplay between both phases and their corresponding moves keeps a high degree of connectivity in the solution, which is enough to obtain a large number of feasible solutions during the search process.

A long-term diversification phase is implemented to complement the diversification phase in the basic procedure. It consists of a re-starting mechanism in which frequency information generated by the search history is mapped onto selection probabilities to guide the construction process. A modification of the constructive algorithm is used to generate diverse solutions with good objective values.

The global tabu search algorithm applies the basic procedure to the solutions generated with the re-starting method, but only tries to improve promising solutions: if the value of the solution is within a certain percentage of the best solution found, then the basic procedure is applied. Otherwise, this solution is rejected and a new one is generated. The procedure finishes when, after a number of iterations, none of the visited solutions improves the best known one.

As the best MRPP tour does not necessarily correspond to the best MRPPTP tour, all the MRPP tours found with the above heuristics and with a cost lower than 1.05 of the cost of the best known MRPP tour are stored. All of them will be the input for the two phases described in the following subsections. The best MRPPTP tour obtained will be the output of the global procedure.

4.2. Obtaining MRPPTP tours

This phase of our heuristic consists of transforming MRPP tours into MRPPTP tours. To describe it we need the following definition:

- Let T be a *MRPPTP* tour and let C be a $v-s$ feasible chain from a to b contained in T (C is a section of T). Then, C is a *non-required section* of T if, when we delete $C\backslash a$ from T, the remaining section of T still traverses each required link of G at least once. A non-required section C will be *maximal* if it is not contained in a longer non-required section of T. Note that if a $v-s$ feasible chain from a to b is a maximal non-required section of a MRPPTP tour, then a and b are required links.

Let T be a MRPP tour. First, T is partitioned into a set of circuits by solving an Assignment Problem at each vertex, in which an arc entering that vertex is assigned to an arc leaving it, as in the Bodin and Kursh [10,11] heuristic (see Section 2). Then, these circuits are joined into a single one by using the CROSSING procedure described in Benavent and Soler [14]. Basically, the procedure consists of constructing an undirected and connected graph H, where each vertex i corresponds to a circuit C_i and $(i, j)\in E(H)$ iff circuits C_i and C_j have at least one vertex in common. If we denote by $a\in C_i$ and $b\in C_j$ two arcs entering at a common vertex, and by a' and b' their successor arcs in C_i and C_j, respectively, the cost associated with edge (i, j) is defined as

$c_{ij} = \min\{p_{[ab']} + p_{[ba']} - p_{[aa']} - p_{[bb']}: \forall a\in C_i, \forall b\in C_j$ being arcs entering at the same node$\}$, where forbidden turns have an associated very high penalty.

Once graph H has been created, a shortest spanning tree (SST) is computed. For each edge (i, j) in the SST solution, their associated circuits C_i and C_j are joined by replacing turns $[a, a']$ and $[b, b']$ by the new turns $[a, b']$ and $[b, a']$, where $a\in C_i$ and $b\in C_j$ are the arcs for which the minimum was obtained when computing c_{ij}. If the resulting circuit, say again T, does not violate any forbidden turns, T is a MRPPTP tour. Otherwise, maximal non required sections of T containing forbidden turns are found and replaced by their corresponding shortest feasible chains.

4.3. Improving MRPPTP tours

Given a MRPPTP tour, in this phase we try to improve it. The procedure used here is an extension to the mixed case of algorithm IMPROVE given in Benavent and Soler [14] for the directed case. It is based on the substitution of maximal non-required sections of the tour by shortest feasible chains in what they call 3-interchanges:

Let C be a circuit passing at least three times through a given vertex v. Let a_1, a_2 and a_3 be three arcs entering at vertex v and b_1, b_2 and b_3 three arcs leaving vertex v such that C can be written as $a_1b_1S_1a_2b_2S_2a_3b_3S_3$, where S_1, S_2 and S_3 are sections of C. A 3-*interchange* associated to arcs a_1, a_2 and a_3 consists of substituting the three turns made by C at v, $[a_1, b_1]$, $[a_2, b_2]$ and $[a_3, b_3]$, by turns $[a_1, b_2]$, $[a_2, b_3]$ and $[a_3, b_1]$ if none of these last three turns are forbidden. The result of a 3-interchange is a new circuit that traverses the same set of arcs in a different order: $a_1b_2S_2a_3b_1S_1a_2b_3S_3$. Circuit cost changes into $p_{[a_1b_2]} + p_{[a_2b_3]} + p_{[a_3b_1]} - p_{[a_1b_1]} - p_{[a_2b_2]} - p_{[a_3b_3]}$, are called *cost of the* 3-*interchange*.

The procedure is based on the following two steps. Given a MRPPTP tour T:

Step 1: Find maximal non-required sections in T and replace them with their corresponding shortest feasible chains. Let T be the new MRPPTP tour.

Step 2: For every vertex v visited by T and for every arc a of T entering at v do, if possible, a 3-interchange involving a with negative cost and update T. If no such 3-interchange exists, do any 3-interchange involving a (if one exists).

The idea behind Step 2 is to modify the order of traversing links in T in such a way that new nonrequired sections can be found in Step 1. Then, if any 3-interchange has been done in Step 2, the procedure goes again to Step 1.

Step 1 stops when n' iterations are performed without improving the value of T. Let n be the number of the last iteration in which T was modified. The entire procedure (Steps 1 and 2) is performed for $n + n'$ global iterations. After some computational experiments, n' was fixed to 20. This number has been chosen from among several values ($n' \leqslant 50$) tested in our computational experience. Larger values produced almost the same results but with larger computational times.

5. Computational experiments

The computational experiments have been designed to assess the efficiency of the heuristic procedure described in Section 4 as well as the transformation given in Section 3. Based on this transformation of the MRPPTP into the ATSP two algorithms have been considered. A Branch & Bound for the ATSP due to Fischetti and Toth [19] and the patching heuristic due to Karp [20]. The first one is considered one of the best exact methods published, while the second one has been proven to be very efficient [5]. Both algorithms can be used to solve the MRPPTP by first transforming it into an ATSP, solving it and then transforming the ATSP solution into an MRPPTP solution. In what follows, *Opt* will denote the exact method while $H1$ will represent the heuristic procedure. Similarly, $H2$ will represent the heuristic algorithm described in Section 4.

We have tested the procedures on four sets of random instances. Instances were generated as follows:

First, $|V|$ vertices are randomly selected on the $[0,100]^2$ grid. Two random cyclic permutations (Hamiltonian directed circuits) without common arcs are also generated and added in order to avoid infeasible instances. Then, $|A|$ ($|A| > 2|V|$) randomly generated arcs are added to the graph, and some of them will be randomly selected as required arcs. Finally, $|E|$ edges are added, all of them to be considered as required edges. Link costs will be set equal to the Euclidean distance between its terminal vertices rounded to the nearest integer. Note that instances do not contain non-required edges as any non-required edge can be replaced by two non-required arcs with opposite directions, both with the edge cost.

More specifically, to generate the first set of instances, for each pair ($|V|$, $|A|$) in the set:

{(40, 90), (40, 110), (60, 140), (60, 160), (80, 170), (80, 200), (100, 220), (100, 250), (120, 260), (120, 290), (140, 300), (140, 330), (160, 340), (160, 370), (180, 380), (180, 400), (200, 420), (200, 440)},

three percentages of required arcs, 40, 70 and 100%, have been considered. For each value of ($|V|$, $|A|$, $|A_R|$), we obtained four graphs by adding 10, 20, 30 and 40 edges, respectively. Finally, as in other computational experiments which have been published, we assigned the following turn

Table 1
Results on the first set of random instances

$\|A_R\|$	$\|E\|$	*Ins*	T_{Opt}	T_{H1}	T_{H2}	C_{H1}/C_{H2}	C_{H2}/C_{Opt}
[34, 95[	10	10	2	1	8	1.034	1.018
[95, 140[	10	10	4	2	4	1.026	1.016
[140, 180[	10	10	6	3	102	1.016	1.018
[180, 260[	10	9	8	3	2	1.024	1.003
[260, 330[	10	8	34	4	7	1.021	1.004
[330, 440[	10	7	60	6	3	1.021	1.002
[34, 95[	20	10	17	1	5	1.062	1.020
[95, 140[	20	10	61	2	23	1.062	1.016
[140, 180[	20	10	175	3	66	1.039	1.022
[180, 260[	20	9	146	3	3	1.048	1.005
[260, 330[	20	8	1326	5	7	1.043	1.005
[330, 440[	20	7	2601	7	3	1.052	1.002
[34, 95[	30	10	1556	1	4	1.102	1.015
[95, 140[	30	10	1432	2	17	1.090	1.017
[140, 180[	30	10	>5522	3	39	1.069	1.021
[180, 260[	30	9	>9595	4	3	1.059	1.007
[260, 330[	30	8	$>17\,703$	5	3	1.062	1.006
[330, 440[	30	7	$>25\,470$	8	3	1.049	1.004
[34, 95[	40	10	$>15\,855$	2	4	1.115	1.021
[95, 140[	40	10	$>22\,148$	3	15	1.108	1.013
[140, 180[	40	10	$>27\,534$	4	36	1.086	1.004
[180, 260[	40	9	$>21\,270$	4	3	1.091	1.006
[260, 330[	40	8	$>28\,800$	7	8	1.069	—
[330, 440[	40	7	$>28\,800$	10	3	1.067	—

penalties: 0 for going straight ahead, 1 for turning right, 3 for turning left and all U-turns were considered forbidden. Then, the first set of instances consists of $18 \times 3 \times 4 = 216$ instances.

Table 1 shows the results of the two heuristics and the exact method for each group of instances in the first set according to the number of required arcs ($|A_R|$) and the number of (required) edges ($|E|$). For each group, *Ins* gives the number of instances and T_{Opt}, T_{H1} and T_{H2} give the average run time in seconds (rounded integer) of each procedure on a Pentium II-350 Personal Computer. C_{H1}/C_{H2} gives the average ratio between the solution costs of both heuristics, while C_{H2}/C_{Opt} gives the average ratio between the solution costs of $H2$ and the optimal costs. Symbol " $>$ " in the T_{Opt} column indicates that one or more instances in the corresponding group cannot be optimally solved in 8 h. In this case, the average value C_{H2}/C_{Opt} is computed over the instances with known optimal solution. A value of "28800" in the T_{Opt} column indicates that no instance in the group could be optimally solved, and then there is no C_{H2}/C_{Opt} value.

Results in Table 1 show that the exact procedure is able to solve all the instances in the groups with 10 and 20 required edges, but was unable to solve 23 and 75% of the instances with 30 and 40 edges, respectively, within 8 h of computer time. Procedure $H1$ is clearly inferior to procedure $H2$ in terms of solution quality, but given its low running time, its performance is quite acceptable. The

Table 2
Computational results on instances with 60 required edges

	$\|V\|$	$\|A\|$	$\%RA$	C_{H1}	T_{H1}	C_{H2}	T_{H2}	C_{H1}/C_{H2}
1	80	200	40	11 200	2	10 160	3	1.102
2	80	200	70	14 852	3	12 872	1	1.153
3	80	200	100	16 864	4	15 958	1	1.056
4	100	220	40	10 939	2	9664	4	1.131
5	100	220	70	15 393	3	14 463	2	1.064
6	100	220	100	17 285	4	16 168	1	1.069
7	120	260	40	14 282	3	12 805	11	1.115
8	120	260	70	19 618	4	17 029	2	1.152
9	120	260	100	23 211	5	20 835	2	1.114
10	140	330	40	16 032	5	14 282	17	1.122
11	140	330	70	22 263	6	19 601	3	1.135
12	140	330	100	25 681	9	24 241	3	1.059
13	160	370	40	18 896	6	16 693	28	1.131
14	160	370	70	25 792	8	23 310	3	1.106
15	160	370	100	30 321	11	28 695	3	1.056
16	180	400	40	20 440	6	18 361	33	1.113
17	180	400	60	26 502	9	23 277	3	1.138
18	180	400	100	30 092	12	28 391	4	1.059
19	200	420	40	22 389	7	19 854	76	1.127
20	200	420	70	27 688	10	25 002	4	1.107
21	200	420	100	32 869	14	30 094	4	1.092
Average results					7		10	1.104

behaviour of heuristic $H1$ and the exact method, although very good in instances with a small number of required edges, quickly deteriorates as this number increases. On the other hand, the performance of $H2$ seems not to depend on the number of required edges.

In the following experiments we compare the performance of both heuristics on denser graphs relative to the number of required edges. Then, optimal solutions are not known and we only report the run times and the ratio between heuristics. Three sets of instances are considered, each one with 60, 80 and 100 required edges, reported in Tables 2–4, respectively. Each table contains the results of 21 instances, where $|V|$ ranges from 80 up to 200 and $|A|$ from 200 up to 420. The percentage of required arcs ($\%RA$) and turn penalties are set as in the first set of instances.

Table 2 shows that $H2$ clearly outperforms $H1$ since in all the instances C_{H1}/C_{H2} is greater than 1 and on average is equal to 1.104 (which is equivalent to a 10.4 percentage deviation). $H1$ is faster than $H2$, but both algorithms present low running times (7 s on average for $H1$ vs. 10 s for $H2$).

Tables 3 and 4 present similar results which are summarized in Table 5. It is worth mentioning that as the number of required edges increases, optimal and $H1$ solutions are harder to obtain in terms of computer time. In fact, on larger instances relative to the number of required edges, running times of $H1$ are even greater than those of $H2$. These experiments confirm that $H2$ is quite robust since its performance does not vary across all the instances under consideration.

Table 3
Computational results on instances with 80 required edges

	$\|V\|$	$\|A\|$	$\%RA$	C_{H1}	T_{H1}	C_{H2}	T_{H2}	C_{H1}/C_{H2}
1	80	200	40	11 376	3	9560	1	1.189
2	80	200	70	14 570	4	13 179	1	1.105
3	80	200	100	18 326	5	17 698	2	1.097
4	100	220	40	13 524	3	11 193	6	1.208
5	100	220	70	15 564	4	13 957	2	1.115
6	100	220	100	18 758	5	16 958	2	1.106
7	120	260	40	16 047	4	13 363	11	1.200
8	120	260	70	19 432	6	17 738	2	1.095
9	120	260	100	23 825	7	21 913	2	1.087
10	140	330	40	17 147	6	14 975	14	1.145
11	140	330	70	22 210	8	19 837	3	1.119
12	140	330	100	28 801	11	25 436	3	1.132
13	160	370	40	20 664	7	17 628	39	1.172
14	160	370	70	26 541	10	24 044	3	1.103
15	160	370	100	32 344	14	29 687	4	1.089
16	180	400	40	20 079	8	18 006	36	1.115
17	180	400	60	26 375	12	23 570	4	1.119
18	180	400	100	32 497	15	29 231	5	1.111
19	200	420	40	22 519	9	18 903	41	1.191
20	200	420	70	28 537	13	25 505	18	1.118
21	200	420	100	33 702	17	30 687	5	1.098
Average results					9		10	1.129

6. Conclusions

In this paper, we have focused on a new problem, the MRPPTP, which generalizes many of the single vehicle Arc Routing Problems already studied, by introducing turn penalties and forbidden turns on a mixed graph. This problem has a practical importance, especially when routes have to be designed inside a city in order to solve real applications.

Although it has been proved here that even a very much simpler case of the MRPPTP is NP-hard (Section 3), some insights into its resolution have also been obtained. From a theoretical point of view, we show that it is possible to solve the MRPPTP, both optimally and heuristically, by transforming it into an ATSP. On the other hand, we have also presented a specific method to solve this problem based on a tabu search metaheuristic.

The performance of the procedures has been assessed using 279 problem instances of several sizes and types. The comparison favors the proposed tabu search implementation compared to the heuristic based on the ATSP transformation. This last heuristic and the exact method based on the branch and bound by Fischetti and Toth [19] seem to be adequate on instances with few required edges. The tabu search heuristic has been shown to be robust in terms of solution quality and computational effort (a deviation from optimality of about 1% on average for problems where the optimum could be found).

Table 4
Computational results on instances with 100 required edges

	$\|V\|$	$\|A\|$	$\%RA$	C_{H1}	T_{H1}	C_{H2}	T_{H2}	C_{H1}/C_{H2}
1	80	200	40	13 628	4	11 737	2	1.161
2	80	200	70	16 958	5	14 988	2	1.131
3	80	200	100	19 076	6	17 487	2	1.090
4	100	220	40	13 767	5	11 405	5	1.207
5	100	220	70	17 790	6	15 376	2	1.156
6	100	220	100	20 291	7	18 232	2	1.112
7	120	260	40	15 402	6	13 044	13	1.180
8	120	260	70	20 628	7	18 119	7	1.138
9	120	260	100	24 607	9	22 269	3	1.104
10	140	330	40	18 070	8	15 007	18	1.204
11	140	330	70	23 703	11	21 297	4	1.112
12	140	330	100	28 973	14	25 772	4	1.124
13	160	370	40	22 879	9	18 840	22	1.214
14	160	370	70	28 152	13	24 989	4	1.126
15	160	370	100	33 589	17	30 773	5	1.091
16	180	400	40	21 981	11	18 615	39	1.180
17	180	400	60	27 570	15	23 913	4	1.152
18	180	400	100	32 190	19	29 004	5	1.109
19	200	420	40	24 346	11	20 105	47	1.210
20	200	420	70	31 428	16	26 617	5	1.180
21	200	420	100	36 195	21	32 683	6	1.107
Average results					11		10	1.147

Table 5
Average results on the number of required edges

$\|E\|$	*Ins*	T_{Opt}	T_{H1}	T_{H2}	C_{H1}/C_{H2}	C_{H2}/C_{Opt}
10	54	17	3	23	1.023	1.010
20	54	605	3	19	1.051	1.012
30	54	>9100	4	12	1.073	1.014
40	54	$>23\,682$	4	12	1.091	1.013
60	21	—	7	10	1.104	—
80	21	—	9	10	1.129	—
100	21	—	11	10	1.147	—

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Angel Corberán is a Professor in the Department of Statistics and Operations Research at the University of Valencia in Spain. His areas of research include polyhedral combinatorics, routing problems, production planning and the development of heuristic procedures in combinatorial optimization. He is Associate Editor of TOP (Journal of the Spanish OR Society).

Rafael Martí is a Professor in the Department of Statistics and Operations Research at the University of Valencia in Spain. His areas of research include graph drawing, routing problems and metaheuristics in combinatorial optimization.

Eulalia Martínez took her master degree on Mathematics at the Polytechnic University of Valencia (Spain) in 1998, where she is an Assistant Professor in the Department of Applied Mathematics. Now she is working in her Ph.D. thesis on arc routing problems.

David Soler received his Ph.D. degree in Mathematics from the University of Valencia in 1995. He is working on combinatorial optimization, mainly in vehicle routing. He is a Professor in the Department of Applied Mathematics at the Polytechnic University of Valencia.